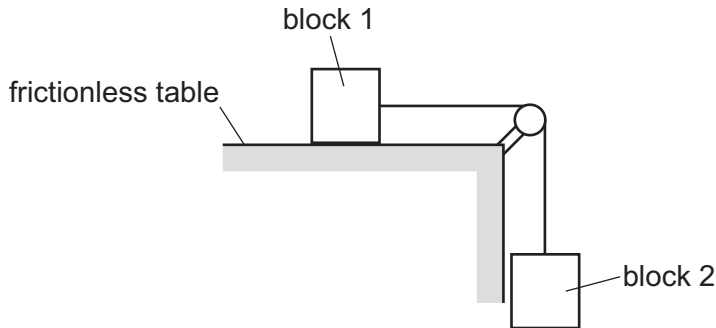


- 10** Block 1 of mass 3.0 kg is on a horizontal frictionless table. A rope with negligible mass is attached to the block.

The rope passes over a frictionless pulley.

Block 2 of mass 1.0 kg is suspended from the other end of the rope, as shown.



The blocks accelerate.

The acceleration of free fall is g . The unit of g is m s^{-2} and the tension in the rope has the unit N.

What is the tension in the rope in terms of g ?

A $\frac{g}{4}$

B $\frac{g}{2}$

C $\frac{3g}{4}$

D g